

SHISANJIANFANG, China

WMO: 514950

Lat: 43.2124N

Lon: 91.7424E

Elev: 2402

StdP: 13.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	4.4	-19.0	2.1	16.1	-14.8	2.7	17.9	51.4	24.4	47.1	21.5	4.0	0	0.851

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	101.1	63.4	98.3	62.6	95.7	61.7	65.5	91.9	64.4	90.8	63.4	89.7	13.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	76.6	75.2	54.4	68.9	76.4	51.8	62.5	77.6	31.7	91.2	30.7	90.7	30.0	89.6	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
54.3	49.0	44.3	-1.6	105.6	4.8	2.5	-5.0	107.5	-7.9	109.0	-10.6	110.4	-14.1	112.2		
			-2.7	67.6	4.6	2.3	-6.0	69.2	-8.7	70.6	-11.2	71.9	-14.5	73.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	17.8	29.2	44.1	58.7	70.2	80.6	84.3	82.1	70.9	55.0	37.7	21.5	(6)
(7)		DBStd	24.31	6.06	7.36	10.12	9.68	8.06	6.81	6.24	6.51	7.66	8.57	8.60	7.09	(7)
(8)		HDD50	3170	999	584	238	37	1	0	0	1	50	376	884	(8)	
(9)		HDD65	5914	1464	1004	649	229	43	1	0	32	323	819	1349	(9)	
(10)		CDD50	4797	0	0	54	297	627	918	1065	994	630	205	7	0	(10)
(11)		CDD65	2065	0	0	0	39	204	469	600	530	210	13	0	0	(11)
(12)		CDH74	24854	0	0	2	333	1985	5793	7975	6728	1959	79	0	0	(12)
(13)		CDH80	12674	0	0	0	68	767	2981	4515	3617	718	8	0	0	(13)
(14)	Wind	WSAvg	17.2	8.6	13.8	18.2	21.0	22.1	22.8	21.1	18.4	15.7	12.8	9.4	(14)	
(15)	Precipitation	PrecAvg	0.7	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.1	0.0	0.0	0.0	(15)	
(16)		PrecMax	1.5	0.2	0.0	0.0	0.5	0.2	0.7	0.9	0.8	0.4	0.1	0.2	0.1	(16)
(17)		PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.4	0.0	0.0	0.0	0.1	0.1	0.2	0.3	0.2	0.1	0.0	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	52.2	72.7	86.4	95.1	102.5	105.9	103.1	94.4	80.8	62.9	41.1	(19)
(20)		MCWB	28.9	37.1	47.8	55.7	59.1	62.9	65.5	63.8	60.0	53.9	44.8	31.8	(20)	
(21)		2%	DB	32.5	47.8	68.0	81.1	91.4	98.5	102.1	99.9	90.7	76.0	57.3	37.4	(21)
(22)		MCWB	25.5	34.6	45.7	52.8	57.8	61.9	64.2	62.6	58.7	51.3	41.9	29.6	(22)	
(23)		5%	DB	29.8	44.4	63.8	77.7	88.0	95.7	99.2	97.0	87.6	72.3	53.8	34.4	(23)
(24)		MCWB	23.5	32.6	44.1	51.2	56.7	61.2	63.4	62.0	57.5	49.8	40.1	27.2	(24)	
(25)		10%	DB	27.2	41.3	59.6	74.0	84.5	92.9	96.2	94.2	84.2	68.9	50.4	31.8	(25)
(26)		MCWB	21.6	30.9	42.1	49.7	55.3	60.4	62.5	61.2	56.3	48.2	38.5	25.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.3	37.7	48.8	56.4	60.9	65.5	67.8	66.7	61.8	54.7	45.6	32.5	(27)
(28)		MCDWB	36.1	50.9	69.1	82.4	91.9	90.3	97.3	91.3	89.4	78.4	60.5	40.0	(28)	
(29)		2%	WB	25.8	35.0	46.7	54.1	59.0	64.1	66.1	64.7	59.9	52.5	42.7	29.9	(29)
(30)		MCDWB	32.1	46.7	66.0	78.0	87.7	90.1	93.3	92.1	86.2	73.7	55.7	37.2	(30)	
(31)		5%	WB	23.7	33.1	44.7	52.3	57.5	63.0	64.9	63.5	58.5	50.6	40.6	27.6	(31)
(32)		MCDWB	29.4	44.1	62.1	74.4	84.1	89.0	91.1	91.2	84.3	70.6	52.7	34.1	(32)	
(33)		10%	WB	21.7	30.9	42.7	50.7	56.1	61.8	63.9	62.4	57.1	48.7	38.6	25.6	(33)
(34)		MCDWB	27.0	41.0	59.0	72.0	81.8	88.0	89.7	89.7	82.8	67.4	49.9	31.5	(34)	
(35)	Mean Daily Temperature Range		MDBR	15.4	17.8	19.2	19.2	18.4	17.4	17.3	18.1	19.2	18.8	15.9	14.2	(35)
(36)		5% DB	MCDBR	16.9	20.3	22.1	22.8	21.8	19.7	19.9	21.0	21.6	21.8	19.8	15.4	(36)
(37)			MCWBR	12.3	12.9	11.1	10.5	9.3	7.1	6.8	7.6	8.6	10.3	11.3	10.7	(37)
(38)			MCDBR	16.2	19.4	20.2	20.1	18.9	17.2	17.2	18.2	19.6	19.9	18.5	15.1	(38)
(39)		5% WB	MCWBR	12.0	12.4	10.3	9.3	8.3	6.1	5.8	6.6	8.1	9.4	10.8	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.373	0.416	0.461	0.463	0.469	0.458	0.432	0.413	0.396	0.363	0.334	(40)	
(41)		taud	2.120	2.061	1.968	1.916	1.966	2.029	2.102	2.155	2.126	2.081	2.112	2.130	(41)	
(42)		Ebn at Noon	241	252	256	255	259	257	258	261	255	240	227	228	(42)	
(43)		Edh at Noon	32	41	51	58	57	53	49	45	43	39	32	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	695	1026	1483	1879	2213	2255	2145	1993	1695	1224	774	592	(44)	
(45)		RadStd	40	50	58	111	96	142	85	98	62	41	45	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+2.14	+3.62	N/A	+1.69	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+1.01	N/A	N/A	+0.97	N/A	-1.40	N/A	N/A	+263	+182

Nomenclature: See separate page